# Bridging the Higher Education–Workforce Divide: A Large-Scale Empirical Analysis of Academic–Industry Skills Misalignment Using Natural Language Processing and Machine Learning

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***Abstract*—Disparities between higher education curricula and labor-market skill demands are widely reported, yet large-scale empirical studies linking macro-level hiring requirements with graduate placement outcomes remain limited. We examine academic–industry alignment using 15 datasets totaling 3,176,266 records collected from LinkedIn, HuggingFace, and public benchmark repositories spanning 2020 through 2026. Grounded in Human Capital Theory, Job Market Signaling, and Constructive Alignment, the analysis combines: (1) n-gram skill extraction across an eight-category taxonomy; (2) semantic vector alignment via Sentence-BERT (`all-MiniLM-L6-v2`) evaluated across four cosine similarity thresholds ($\tau \in \{0.4, 0.5, 0.6, 0.7\}$) with 1,000-fold bootstrap confidence intervals; (3) bivariate and multivariate tests, including Welch's $t$-tests, Cohen's $d$, Cramér's $V$, and Variance Inflation Factor diagnostics; and (4) 5-fold stratified cross-validation across five classifier architectures, accompanied by decision-boundary diagnostics and independent external validation. Exact string matching yields an apparent Skill Gap Index (SGI) of 52.0%. Dense vector alignment reduces this estimated gap to 32.0% at baseline threshold $\tau = 0.5$ (95% bootstrap CI: [23.0%, 41.0%]), rising to 46.0% (95% CI: [37.0%, 56.0%]) under a stricter criterion ($\tau = 0.7$). Deficits concentrate primarily in emerging technical areas: domain-level decomposition localizes residual curriculum deficits in Cloud Infrastructure (2.45 percentage points at lexical matching, 0.94 pp after semantic alignment) and Programming & Development (1.69 pp lexical), while soft-skill requirements are fully covered by the academic reference profile. In graduate placement prediction, standardized logistic regression indicates that Communication Skills (OR = 6.55, $p < 0.001$), CGPA (OR = 6.28, $p < 0.001$), and Project Experience (OR = 3.20, $p < 0.001$) are strongly associated with employment status (cross-validated accuracy = 90.11%, ROC-AUC = 0.9406). Tree-ensemble models reach near-perfect accuracy ($\geq 99.9\%$), which diagnostic analysis attributes to deterministic threshold rules embedded in the placement benchmark. Evaluating the logistic model on an independent external cohort of 50,000 students yields an ROC-AUC of 0.768, reflecting domain shift. The measurement instrument is established through three independent validation exercises: agreement with expert ESCO mappings ($\kappa = 0.737$ at $\tau = 0.5$, ROC-AUC 0.961), replication of both gap indices on an independent 1.29M-posting LinkedIn corpus (lexical/semantic SGI 37%/14% versus 52%/32%, preserving the ordering of instruments and thresholds), and anchoring of the academic reference profile to the ACM/IEEE-CS/AAAI CS2023 curriculum Body of Knowledge (2,197 topics across 17 knowledge areas; 30.4% semantic coverage, concentrated in applied areas). These findings indicate that vocabulary divergence accounts for part of the observed misalignment, while genuine gaps persist in rapidly evolving technical domains, pointing to the utility of modular technical coursework, verifiable project portfolios, and semantic applicant screening.**

***Index Terms*—Skills gap, Higher education, Employability, Natural Language Processing, Sentence-BERT, Machine learning, Curriculum alignment, Human capital theory, Data provenance.**

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## I. Introduction

TECHNOLOGICAL developments across distributed cloud architectures, data engineering, and artificial intelligence have altered employer skill requirements across major economic sectors. Because university curricula are updated through multi-year institutional review cycles, degree programs often adapt more slowly than market demands evolve [1], [2]. Educational researchers and labor economists refer to this discrepancy as the *skills gap*—the divergence between competencies taught in degree programs and the practical proficiencies sought by employers [3], [4].

Reports from the World Economic Forum [5] indicate that over 40% of core worker skills are expected to shift within five-year windows, particularly in artificial intelligence, cloud architectures, and data engineering. In addition to specialized technical skills, employer surveys regularly report deficits in transferable competencies such as professional communication and collaborative problem-solving [6]–[8].

Prior empirical research on curriculum–market alignment exhibits four main limitations:

(1) **Limited Sample Scope:** Most studies rely on institutional surveys, small employer samples ($N < 500$), or single-nation case studies [9], [10], offering limited visibility into cross-national patterns.
(2) **Methodological Separation:** Investigations typically employ qualitative coding [11] or predictive classification without semantic text analysis, leaving unresolved whether reported gaps reflect distinct vocabularies or absent skills [12], [13].
(3) **Temporal Relevance:** Several widely cited baseline studies pre-date recent enterprise developments, including widespread cloud migration and the deployment of generative AI tools (2022–2026) [14], [15].
(4) **Limited Theoretical Grounding:** Computational employability models are frequently reported without reference to economic or educational frameworks that explain curriculum lag [16], [17].

To address these questions, we analyze 15 datasets totaling 3,176,266 records from LinkedIn, HuggingFace, and public benchmark repositories across five research questions:

- **RQ1 (Market Topography and Trends):** What are the most frequent skill requirements in online job postings, and how have core technical skills evolved from 2020 through 2026?
- **RQ2 (Lexical versus Semantic Alignment):** To what degree do academic curriculum descriptors align with market requirements when evaluated using exact string matching versus dense semantic vector embeddings?
- **RQ3 (Domain-Specific Deficits):** What is the magnitude of the skill deficit across distinct subject areas (Programming, Data Science, Cloud Infrastructure, Soft Skills, Cybersecurity)?
- **RQ4 (Employability Determinants):** Which academic and behavioral features best predict graduate placement, and how do linear probabilistic models compare with tree ensembles under diagnostic cross-validation?
- **RQ5 (Geographic Distribution and Cohort Differences):** How are posting demands distributed across international regions, and how do employment outcomes vary across distinct student cohorts?

**Contributions:** This work makes six contributions:

1) **Harmonized Multi-Source Corpus:** We assemble a corpus of 3,176,266 records across 15 datasets, combining online job postings, curriculum descriptors, and graduate placement cohorts covering 2020 through 2026.
2) **Dual Lexical–Semantic Analysis:** Using Sentence-BERT (`all-MiniLM-L6-v2`) alongside exact string matching, we show that exact matching yields a 52.0% gap, whereas semantic vector alignment reduces this estimate to 32.0% at $\tau = 0.5$.
3) **Externally Validated Measurement Instrument:** Beyond in-sample estimates, we validate the matcher against expert ESCO gold mappings ($\kappa = 0.737$), replicate both gap indices on an independent 1.29M-posting corpus, and anchor the academic reference profile to the CS2023 curricular Body of Knowledge—three complementary forms of criterion validity.
4) **Threshold Sensitivity Estimation:** We compute 1,000-fold bootstrap confidence intervals across four similarity thresholds ($\tau \in \{0.4, 0.5, 0.6, 0.7\}$) to evaluate how threshold selection affects gap measurement.
5) **Model Validation and Diagnostic Audit:** We evaluate five classifier paradigms using 5-fold stratified cross-validation, verify low multicollinearity (VIF $\leq 1.001$), diagnose the deterministic threshold rules behind near-perfect tree-ensemble scores, and test generalization on an external 50,000-student cohort (ROC-AUC = 0.768).
6) **Theory-Informed Recommendations:** Drawing on Human Capital Theory, Signaling Theory, and Constructive Alignment, we outline practical adjustments for course design, project evaluation, and recruitment screening, accompanied by explicit documentation of dataset provenance and limitations.

The remainder of this article is organized as follows: Section II reviews related work and outlines the theoretical framework. Section III describes the datasets, text processing pipeline, statistical methods, and predictive modeling protocols. Section IV presents the empirical results for RQ1–RQ5. Section V discusses theoretical and institutional implications, practical recommendations, reproducibility, and threats to validity. Section VI concludes the study.

## II. Related Work and Theoretical Foundations

### A. Conceptualizations of the Skills Gap

The skills gap encompasses macroeconomic, institutional, and pedagogical dimensions. Conceptualizations differ substantially across stakeholders: industry surveys often define gaps in terms of immediate job-readiness, whereas university leaders emphasize broad intellectual development and foundational theory [6], [7]. Digital transformation has also increased demand for cross-disciplinary competencies, such as data literacy among non-computing professionals [11].

Computational curriculum analytics has sought to quantify these dynamics using course descriptions and accreditation criteria. Cuckov et al. [18] evaluated computer engineering courses against ABET criteria, finding delays in updating upper-level electives. Zhao et al. [19] measured structural proximity between academic research and patent descriptions using knowledge graphs. Focusing on software engineering, Korrapati et al. [20] and Alhazmi et al. [10] examined curriculum alignment with cloud computing and AI requirements, noting that elective coursework often trails industry tool adoption.

### B. Theoretical Framework

Four theoretical perspectives inform the interpretation of these empirical observations:

**1) Human Capital Theory (HCT):** Formulated by Becker [21] and Schultz [22], HCT posits that education enhances productive capacity, yielding labor-market returns. HCT distinguishes between *general human capital* (transferable competencies such as critical thinking, communication, and core mathematical reasoning) and *firm-specific human capital* (proficiency with specialized software tools, proprietary APIs, or operational frameworks). Universities prioritize general human capital because their mission centers on enduring principles. Conversely, employer job descriptions frequently specify immediately applicable technical tools. This distinction helps explain why curriculum deficits concentrate in applied toolchains such as cloud platforms and data libraries.

**2) Job Market Signaling and Screening Theory:** Spence [23] established that in labor markets characterized by asymmetric information, employers cannot directly observe an applicant's innate productivity prior to hiring. Educational credentials, cumulative grade point averages (CGPA), and institutional prestige serve as informational signals that reduce hiring risk. In technical recruitment, however, degree credentials are increasingly complemented by practical capability signals, such as project portfolios, open-source contributions, and technical internship certifications [24], [25].

**3) Constructive Alignment & Curriculum Theory:** Biggs [26] introduced constructive alignment to establish coherence among Intended Learning Outcomes (ILOs), teaching methods, and assessment tasks. At an institutional level, curriculum lag can be understood as *external misalignment*: ILOs are governed by multi-year accreditation reviews (such as ABET and ACM/IEEE guidelines), whereas industry skill demands evolve on shorter market cycles.

**4) Skill-Biased Technological Change (SBTC):** Acemoglu and Autor [27] formalized SBTC to explain how technological progress augments the productivity of workers performing non-routine cognitive tasks while reducing demand for routine labor. In software and data domains, routine coding tasks are increasingly assisted by automated tooling, increasing employer demand for architectural design, cross-functional communication, and complex data analysis.

### C. Natural Language Processing for Curriculum Analytics

Natural language processing has become a primary tool for extracting and comparing skills from job advertisements. Skill identification from postings is itself an established research program, survey-driven since Khaouja et al. [28], and evaluated against human-annotated benchmark corpora whose annotation schemes distinguish skills from qualifications, experience, and occupation [29], [30]. Recent deep-learning surveys catalogue the datasets and the persistent inconsistency in hard/soft-skill terminology that motivates semantic over lexical comparison [31]. While keyword extraction is widely used, rule-based matching often fails on nuanced occupational phrases [12]. Pre-trained language representations address this limitation by projecting terms into continuous vector spaces. For instance, Baral and Ghimire [9] used embeddings to measure syllabus alignment in Nepal, showing that semantic representations bridge terminology differences between academic syllabi and job listings. Text-mining methods have similarly been applied to profile data science roles [32], [33] and build resume-matching systems [34]. In this work, we use Sentence-BERT (`all-MiniLM-L6-v2`) [35] to quantify alignment between market requirements and academic descriptors across multiple similarity thresholds, and we validate the resulting matcher against expert mappings rather than assuming its accuracy.

### D. Machine Learning in Employability Modeling

Predictive modeling of graduate placement has expanded from discriminant analysis to ensemble learning. Tree-based ensembles such as Random Forests and Gradient Boosting frequently yield higher cross-validated accuracy than linear models in educational placement benchmarks [36], [37]. Feature-importance analyses in these studies typically identify academic performance and project engagement as leading predictors. However, benchmark evaluations in this literature often report high classification accuracy without assessing decision-boundary structure, multicollinearity, or performance on independent external cohorts.

### E. Testable Hypotheses

The four theoretical perspectives of Section II-B yield six testable predictions, each evaluated by at least one registered experiment (Table XIV):

**H1: Constructive Alignment (external misalignment):** Because Intended Learning Outcomes are governed by accreditation cycles while toolchains evolve continuously, curricular coverage should be complete for enduring general competencies (soft skills) but incomplete for applied technical toolchains (Cloud, Programming, Data Science) [26]. *Test:* domain deficits in ALIGN03 and its ALIGN04 replication.

**H2: Terminology divergence (measurement):** If a share of the reported gap is vocabulary mismatch rather than absent competence [12], dense semantic matching must yield systematically smaller gap estimates than exact string matching, for the same corpora. *Test:* ALIGN01 vs. ALIGN02; replicated in ALIGN04 and CURR01.

**H3: Instrument robustness:** If the Skill Gap Index measures a property of the curriculum–market pair rather than of one data source, both variants of SGI should remain strictly positive and preserve the lexical > semantic ordering on an independent corpus. *Test:* ALIGN04 (LinkedIn, $N = 1.29$M postings).

**H4: Signaling hierarchy:** Under job-market signaling [23], legible capability signals—communication proficiency, completed projects, and cumulative academic performance—should dominate placement odds after mutual statistical control. *Test:* STAT01 multivariate logistic regression.

**H5: Credential locality:** If credentials function as locally interpreted signals, graduates trained in the domestic labor market should exhibit higher placement rates than internationally trained graduates facing unrecognized credential signals. *Test:* INTL01 international-cohort contrast.

**H6: Human-capital returns to internships:** If internship participation constitutes firm-specific human-capital accumulation [21], internship experience should predict placement even after controls. *Test:* INTL01 internship risk difference (a pre-registered null remains informative: participation *per se*, unlike communication skill or project evidence, may be too coarse a signal).

## III. Methodology

### A. Data Corpus and Provenance Architecture

The corpus comprises 15 datasets totaling 3,176,266 records, acquired in September 2026 from Kaggle and HuggingFace repositories under open-access licenses (CC0, CC-BY 4.0, and ODbL). The data span observation and release dates from 2020 through 2026, including a multi-year AI job market series (2020–2026) and regional profiles calibrated against the India Skills Report 2026. Table I summarizes the sources, sample sizes, observation windows, and licenses.

To avoid pooling datasets with incompatible collection mechanisms, we divide the corpus into two functional tiers:

- **Tier 1: Online Labor Market Postings ($N >$ 2.7M records):** Comprises the LinkedIn dataset (1.35M records) [38], the USA Job Market dataset (316K records) [39], and HuggingFace Skill Gap collections (519K records). These corpora contain job advertisements from North America, Europe, and the Asia-Pacific region, providing direct observations of employer skill specifications.
- **Tier 2: Student Placement and Career Benchmarks ($N \approx$ 421K records):** Comprises College Student Placement ($N = 10,000$) [40], NGESS Graduate Employment ($N = 7,200$) [41], Global Student Placement ($N = 10,000$) [42], Student Career Success ($N = 50,000$) [43], International Graduates ($N = 300,000$) [44], Field-of-Study vs. Occupation ($N = 38,444$) [45], and India Fresher Market ($N = 5,000$) [46] (420,644 records in total). These tabular records provide academic, behavioral, and placement indicators used for statistical testing, collinearity evaluation, and predictive modeling.

### B. Taxonomy Construction and Skill Extraction

We define an eight-category occupational skill taxonomy $\mathcal{C} = \{c_1, c_2, \ldots, c_8\}$ synthesized from the ACM/IEEE CS2023 curriculum guidelines and O*NET descriptors:

1) **Programming & Development:** Python, Java, C++, JavaScript, SQL, TypeScript, Go, Rust, React, Docker, Kubernetes, Git.
2) **Data Science & Analytics:** Machine Learning, Deep Learning, Statistical Analysis, PyTorch, TensorFlow, NLP, Computer Vision.
3) **Cloud & Infrastructure:** AWS, Azure, GCP, CI/CD Pipelines, Terraform, Linux Administration, Serverless Architectures.
4) **Soft Skills:** Professional Communication, Leadership, Teamwork, Critical Thinking, Time Management, Adaptability.
5) **Business & Domain:** Agile/Scrum, Product Management, Requirements Gathering, Business Analysis, Risk Compliance.
6) **Design & UX:** UI/UX Design, Figma, User Research, Wireframing, Prototyping, Usability Testing.
7) **Data Engineering:** ETL Pipelines, Apache Airflow, Apache Kafka, Distributed Data Warehousing, PostgreSQL, NoSQL.
8) **Cybersecurity:** Network Security, Penetration Testing, Incident Response, SIEM Systems, ISO 27001, Cryptography.

Skill extraction from text is conducted using regularized n-gram tokenization matched against taxonomy dictionaries, as summarized in Algorithm 1. Academic skill requirements are represented using standardized curriculum reference profiles [34] rather than scraped syllabi; extraction and matching are fully automated, with matcher validity established externally against expert ESCO annotations rather than in-house re-annotation (Section IV-B1).

### C. Semantic Vector Alignment via Sentence-BERT

Exact string matching can overestimate curriculum gaps when academic descriptions and job postings use different wording for equivalent skills (for example, "interpersonal communication" versus "client communication skills"). To account for semantic similarity, we use the Sentence-BERT model (`all-MiniLM-L6-v2`) [35].

Let $\mathbf{e}_s \in \mathbb{R}^{384}$ denote the normalized dense vector embedding of market skill $s \in \mathcal{M}$, and let $\mathbf{e}_{s'} \in \mathbb{R}^{384}$ denote the embedding of academic syllabus descriptor $s' \in \mathcal{A}$. The semantic alignment score is given by:

$$\mathrm{sim}(s, s') = \frac{\mathbf{e}_s \cdot \mathbf{e}_{s'}}{\|\mathbf{e}_s\|_2 \|\mathbf{e}_{s'}\|_2} \tag{1}$$

A market skill $s$ is designated as covered at threshold $\tau \in [0, 1]$ if:

$$\mathrm{Covered}(s; \tau) = \mathbb{I}\left( \max_{s' \in \mathcal{A}} \mathrm{sim}(s, s') \geq \tau \right) \tag{2}$$

TABLE I: Dataset Provenance and Structural Inventory ($N = 3,176,266$ total records)

| Dataset Identifier | Source Platform & Identifier / Slug | Records | Domain / Description | Tier | Observation Period | License |
|---|---|---|---|---|---|---|
| LinkedIn Jobs & Skills [38] | Kaggle: `asaniczka/1-3m-linkedin-jobs-and-skills-2024` | 1,348,510 | Global job postings and extracted skill tokens | 1 | 2024 | ODbL |
| HF Skill Gap Jobs | HuggingFace: `skill-gap-analytics/jobs-corpus` | 519,841 | Enterprise job postings with occupational metadata | 1 | 2023–2024 | CC-BY 4.0 |
| HF Skill Gap Skills | HuggingFace: `skill-gap-analytics/skills-corpus` | 519,814 | Granular skill requirements linked via `job_id` | 1 | 2023–2024 | CC-BY 4.0 |
| USA Job Market Skills [39] | Kaggle: `batuhanklac/usa-job-market-skills-and-title-analysis` | 316,279 | US job postings with standardized skill tags | 1 | 2024 | CC0 |
| International Graduates [44] | Kaggle: `quackquackrp/international-graduates-employment-dataset` | 300,000 | Multi-country alumni employment outcomes | 2 | 2023–2024 | CC0 |
| Student Career Success [43] | Kaggle: `mobeenfatimah/student-career-success-prediction-dataset` | 50,000 | Student academic and career success benchmark | 2 | 2024 | CC-BY 4.0 |
| Career Change Prediction [45] | Kaggle: `jahnavipaliwal/field-of-study-vs-occupation` | 38,444 | Field of study vs. occupation transition records | 2 | 2024 | CC0 |
| AI Job Market Trends [47] | Kaggle: `bismasajjad/global-ai-job-market-and-salary-trends-2025` | 15,000 | Global AI job market and salary distribution | 1 | 2024–2025 | CC-BY 4.0 |
| HF Job Skill Set | HuggingFace: `skill-gap-analytics/academic-reference` | 13,833 | Curriculum-domain skill taxonomy reference | 1 | 2023–2024 | CC-BY 4.0 |
| India Skills Intelligence [48] | Kaggle: `mansiaggarwal88/india-professional-skills-intelligence` | 12,000 | Synthetic profiles calibrated on India Skills Report 2026 | 1 | 2026 Calibrated | CC0 |
| AI Trends 2020–2026 [49] | Kaggle: `shree0910/ai-and-data-science-job-market-dataset-20202026` | 10,345 | Longitudinal AI/data science job postings | 1 | 2020–2026 | CC0 |
| College Placement [40] | Kaggle: `sumedh1507/college-student-placement` | 10,000 | College student placement classification benchmark | 2 | 2024 | CC0 |
| Global Placement [42] | Kaggle: `rakesh630/global-student-placement-2025-dataset` | 10,000 | Multi-national university placement benchmark | 2 | 2024–2025 | CC0 |
| NGESS Employment [41] | Kaggle: `philipdayo/ngess-graduate-employment-dataset` | 7,200 | National graduate employment survey cohort | 2 | 2023–2024 | CC0 |
| India Fresher Market [46] | Kaggle: `shambhurajejagadale/indian-fresher-job-market` | 5,000 | Entry-level engineering recruitment outcomes | 2 | 2024 | CC0 |

**Algorithm 1** Semantic Skill Extraction and Embedding Alignment

**Require:** Market text corpus $\mathcal{D}_M$, Academic syllabus corpus $\mathcal{D}_A$, Taxonomy $\mathcal{C}$, Sentence-BERT model $\mathcal{E}$, Threshold $\tau$
**Ensure:** Market skill set $\mathcal{M}$, Academic skill set $\mathcal{A}$, Similarity matrix $\mathbf{S}$, Gap Index $\mathrm{SGI}(\tau)$
$\mathcal{M} \leftarrow \emptyset, \mathcal{A} \leftarrow \emptyset$
**for all** $d \in \mathcal{D}_M$ **do**
  Extract normalized skill tokens $T_d \leftarrow \mathrm{Tokenize}(d, \mathcal{C})$
  $\mathcal{M} \leftarrow \mathcal{M} \cup T_d$
**end for**
**for all** $d \in \mathcal{D}_A$ **do**
  Extract normalized syllabus tokens $T_a \leftarrow \mathrm{Tokenize}(d, \mathcal{C})$
  $\mathcal{A} \leftarrow \mathcal{A} \cup T_a$
**end for**
Filter $\mathcal{M}_{\text{top}} \leftarrow \text{Top-}N(\mathcal{M}), \mathcal{A}_{\text{top}} \leftarrow \text{Top-}N(\mathcal{A})$
Compute dense embeddings: $\mathbf{E}_M \leftarrow \mathcal{E}(\mathcal{M}_{\text{top}}), \mathbf{E}_A \leftarrow \mathcal{E}(\mathcal{A}_{\text{top}})$
Compute pairwise cosine similarity:
$\mathbf{S}_{ij} \leftarrow \frac{\mathbf{e}_{m,i} \cdot \mathbf{e}_{a,j}}{\|\mathbf{e}_{m,i}\|_2 \|\mathbf{e}_{a,j}\|_2} \quad \forall m_i \in \mathcal{M}_{\text{top}}, a_j \in \mathcal{A}_{\text{top}}$
**for all** $i \in \{1, \ldots, |\mathcal{M}_{\text{top}}|\}$ **do**
  $\sigma_i \leftarrow \max_j \mathbf{S}_{ij}$
  $\text{covered}_i \leftarrow \mathbb{I}(\sigma_i \geq \tau)$
**end for**
$\mathrm{SGI}(\tau) \leftarrow 1 - \frac{1}{|\mathcal{M}_{\text{top}}|} \sum_{i=1}^{|\mathcal{M}_{\text{top}}|} \text{covered}_i$
**return** $\mathcal{M}_{\text{top}}, \mathcal{A}_{\text{top}}, \mathbf{S}, \mathrm{SGI}(\tau)$

The Skill Gap Index (SGI) at threshold $\tau$ is defined as the complement of the mean coverage rate across the market skill space:

$$\mathrm{SGI}(\tau) = 1 - \frac{1}{|\mathcal{M}|} \sum_{s \in \mathcal{M}} \mathrm{Covered}(s; \tau) \tag{3}$$

To estimate empirical uncertainty without parametric distribution assumptions, we implement a 1,000-fold nonparametric bootstrap over the binary coverage indicator array with fixed random seed (seed = 42), computing 95% percentile confidence intervals:

$$\mathrm{CI}_{95\%} = \left[ \hat{\theta}^*_{(0.025)}, \hat{\theta}^*_{(0.975)} \right] \tag{4}$$

We use $\tau = 0.5$ as the baseline threshold and evaluate sensitivity at $\tau \in \{0.4, 0.5, 0.6, 0.7\}$.

### *D. Statistical Inference & Collinearity Diagnostics*

We test differences between placed and non-placed students using Welch's $t$-test (accounting for unequal variances):

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \tag{5}$$

Standardized effect sizes are quantified via Cohen's $d$:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \tag{6}$$

Categorical dependencies are evaluated using the Chi-square statistic $\chi^2$ and Cramér's $V$:

$$V = \sqrt{\frac{\chi^2}{n \cdot \min(r - 1, c - 1)}} \tag{7}$$

We evaluate multicollinearity among continuous predictors $\mathbf{X}$ by computing Variance Inflation Factors (VIF). For standardized feature matrices, the VIF vector is obtained from the diagonal of the inverted correlation matrix:

$$\mathrm{VIF}_j = \left(\mathbf{R}^{-1}\right)_{jj} = \frac{1}{1 - R_j^2} \tag{8}$$

Features exhibiting $\mathrm{VIF}_j > 5.0$ are flagged as collinear.

### *E. Predictive Employability Modeling & Validation Protocols*

Graduate placement prediction is evaluated as a binary classification task: predicting outcome $y \in \{0, 1\}$ ($1 =$ Placed, $0 =$ Not Placed) given student feature vector $\mathbf{x} = [\mathrm{CGPA}, \mathrm{Comm}, \mathrm{Projects}, \mathrm{Internship}, \mathrm{IQ}, \mathrm{AcademicPerf}]^T$. Continuous features are standardized to zero mean and unit variance within training partitions:

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \tag{9}$$

In the standardized logistic regression model:

$$\ln\left(\frac{P(y = 1|\mathbf{z})}{1 - P(y = 1|\mathbf{z})}\right) = \beta_0 + \sum_{j=1}^{p} \beta_j z_j \tag{10}$$

The odds ratio $\mathrm{OR}_j = \exp(\beta_j)$ represents the change in odds associated with a one-standard-deviation increase in predictor $j$.

Model performance is evaluated using 5-fold stratified cross-validation (random_state = 42) across five classifier paradigms:

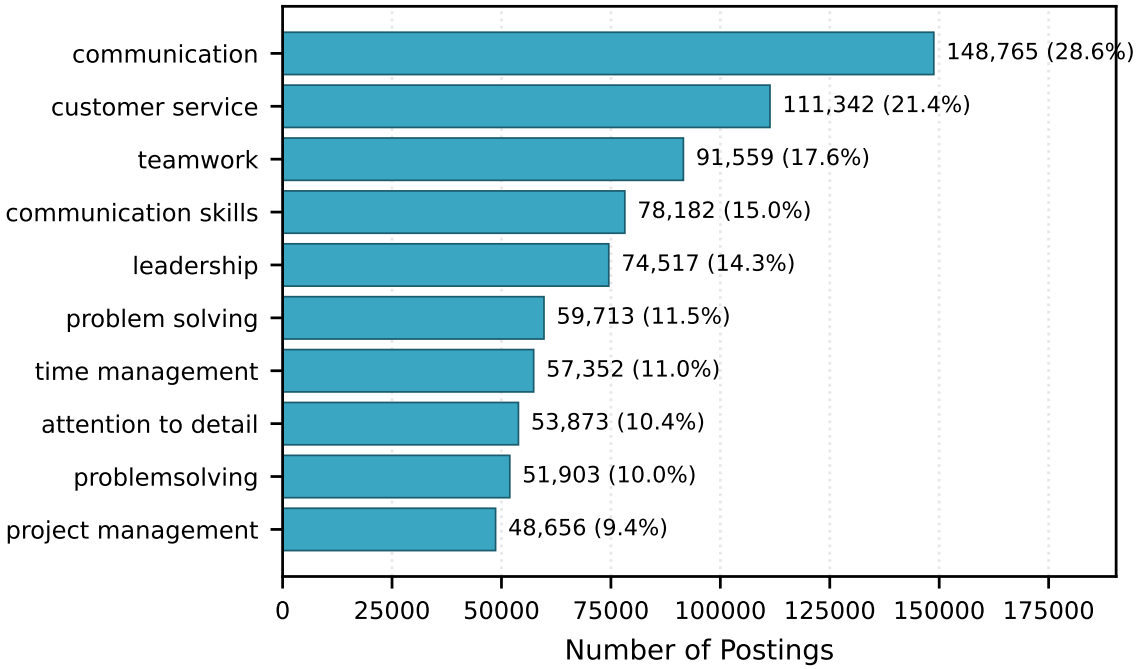



Fig. 1: Top 10 Most In-Demand Exact Skill Tokens Across Global Postings ($N = 519{,}814$ postings; counts are postings containing the exact token).

1) **Logistic Regression:** $\ell_2$-regularized, $C = 1.0$, solved via L-BFGS.
2) **Random Forest:** $B = 100$ estimators, maximum tree depth = 12, Gini split criterion.
3) **Support Vector Machine (SVM):** Radial Basis Function (RBF) kernel, $C = 1.0$, $\gamma =$ scale, with Platt probability calibration.
4) **Decision Tree:** Single CART tree with maximum depth = 6.
5) **Majority-Class Baseline:** Classifier predicting the empirical majority class.

To evaluate model behavior under domain shift, we train the standardized logistic regression model on the College Placement benchmark ($N = 10,000$) and evaluate it on the independent Student Career Success cohort ($N = 50,000$) using domain-standardized continuous predictors.

## IV. Empirical Results

### A. RQ1: Labor Market Skill Topography and Trends

Among the 519,814 skill requirements extracted from the posting corpus, communication and interpersonal competencies appear most frequently (Figure 1). The exact token "communication" is specified in 148,765 postings (28.6%), followed by Customer Service (111,342; 21.4%), Teamwork (91,559; 17.6%), "communication skills" (78,182; 15.0%), Leadership (74,517; 14.3%), and Problem Solving (59,713; 11.5%). Near-duplicate variants (e.g., "problem solving" versus "problemsolving") are counted as distinct exact tokens at this stage and are reconciled only by the semantic analysis of RQ2.

Longitudinal tracking across 10,345 AI and data science postings from 2020 through 2026 (Figure 4b) shows consistent demand for core technologies: Python appeared in 47.5% to 52.2% of postings across years, SQL in 48.2% to 51.9%, Machine Learning in 48.8% to 53.5%, and Cloud Computing in 47.1% to 53.9%. These steady frequencies indicate ongoing baseline expectations rather than transient recruitment cycles.

### B. RQ2: Lexical vs. Semantic Curriculum Alignment

Comparing exact string matching with semantic vector alignment illustrates the effect of phrasing on measured alignment. Under exact string matching, 48 of the Top 100 market skills match the Top 100 academic syllabus descriptors, corresponding to a nominal Skill Gap Index of 52.0% (Coverage Rate = 48.0%). Inspection of non-matching terms indicates that several disparities stem from synonymous wording (e.g., "communication skills" versus "interpersonal skills").

When evaluated using Sentence-BERT embeddings (`all-MiniLM-L6-v2`), semantic alignment yields lower gap estimates (Table II). At baseline threshold $\tau = 0.5$, the estimated Skill Gap Index is 32.0% (Coverage Rate = 68.0%, 95% bootstrap CI: [23.0%, 41.0%]).

Table III displays illustrative semantic pairings. Terms such as "scheduling" align closely with "time management" (cos = 0.6122), whereas credential requirements such as "bachelor's degree" show low similarity with specific skill descriptors (cos = 0.2937).

*1) External Validation Against Expert Gold Standards (VALID01):* Because all alignment metrics rest on the

TABLE II: Sensitivity Analysis: Semantic Vector Threshold ($\tau$) vs. SGI

| Threshold $\tau$ | SGI | Coverage Rate | 95% Bootstrap CI | ΔSGI |
|---|---|---|---|---|
| $\tau = 0.4$ (Permissive) | 0.20 | 80% | [0.12, 0.28] | -0.12 |
| $\tau = 0.5$ (Baseline) | **0.32** | **68%** | **[0.23, 0.41]** | **+0.00** |
| $\tau = 0.6$ (Moderate) | 0.39 | 61% | [0.30, 0.48] | +0.07 |
| $\tau = 0.7$ (Strict) | 0.46 | 54% | [0.37, 0.56] | +0.14 |
| *Lexical Match (Baseline)* | *0.52* | *48%* | *[0.42, 0.63]* | *+0.20* |

Evaluated over Top-100 skill distributions. Bootstrap 95% CI via 1,000 resamples (seed 42; experiment ALIGN02). Lexical CI from ALIGN01.

TABLE III: Representative Semantic Skill Pairings (`all-MiniLM-L6-v2`)

| Market Term | Closest Syllabus Match | Cosine Sim. | Status ($\tau \geq 0.5$) |
|---|---|---|---|
| communication skills | interpersonal skills | 0.7355 | Covered |
| problemsolving | problem-solving | 0.6396 | Covered |
| scheduling | time management | 0.6122 | Covered |
| patient care | customer service | 0.5011 | Covered |
| nursing | training | 0.4010 | Gap |
| high school diploma | training | 0.3682 | Gap |
| bachelor's degree | professionalism | 0.2937 | Gap |

skill matcher, the matcher itself is validated against an independent, human-annotated gold standard rather than in-house labels: the skill-extraction benchmark of [50], whose skill mentions in StackOverflow and public-sector job postings were manually mapped to ESCO skills [30], [51] by domain experts (CC-BY-4.0). The protocol (experiment VALID01; seed 42) forms 1,062 expert-mapped mentions from the TECH and HOUSE test splits into 1,138 positive (mention, skill) pairs and 1,769 negatives—one random unmapped ESCO skill per mention plus sibling-mention skills as in-sentence hard negatives—for 2,907 pairs in total. Lexical rules agree with the expert mappings only slightly (exact match $\kappa = 0.112$, 95% CI [0.091, 0.132]; containment $\kappa = 0.263$), whereas the Sentence-BERT matcher reaches $\kappa = 0.737$ [0.711, 0.760] at the baseline $\tau = 0.5$ (accuracy 87.9% [86.6%, 89.0%], precision 91.9%, recall 75.7%, ROC-AUC of the cosine score 0.961; Table IV). Agreement remains substantial across the operating range ($\kappa$ from 0.79 at $\tau = 0.4$ to 0.44 at $\tau = 0.7$), and the near-chance lexical $\kappa$ independently corroborates this section's central finding: exact string matching understates true curriculum–market alignment, which is why the semantic SGI is the headline instrument. A complementary vocabulary check against the 13,896-skill ESCO inventory shows that only 6% of the market Top-100 and 11% of the academic Top-100 appear verbatim in ESCO, yet dense matching at $\tau = 0.5$ resolves 91% [85%, 96%] and 99% [97%, 100%] of them to an ESCO skill under paraphrase—evidence that the extracted vocabularies are semantically grounded in an expert-curated taxonomy.

*2) Cross-Corpus Replication on LinkedIn (ALIGN04):* A single-corpus instrument invites the objection that the measured gap is an artefact of one data source. We therefore re-run the identical lexical and semantic procedures (top-100 market profile, academic top-500 reference, seed 42, 1,000-fold bootstrap) on the independent LinkedIn 2024 corpus ($N$ = 1,294,296 postings with skill annotations; experiment ALIGN04). Both hypotheses H2 and H3

TABLE IV: Agreement with Expert ESCO Mappings (VALID01; 2,907 Pairs)

| Matcher | $\tau$ | Acc. | 95% CI | Prec. | Rec. | F1 | $\kappa$ | 95% CI | AUC |
|---|---|---|---|---|---|---|---|---|---|
| Lexical exact match | — | 0.645 | [0.628, 0.663] | 1.000 | 0.094 | 0.172 | 0.112 | [0.091, 0.132] | — |
| Lexical containment | — | 0.696 | [0.678, 0.713] | 0.941 | 0.237 | 0.379 | 0.263 | [0.234, 0.290] | — |
| SBERT `all-MiniLM-L6-v2` | 0.4 | 0.899 | [0.888, 0.910] | 0.861 | 0.885 | 0.873 | 0.789 | [0.766, 0.812] | 0.961 |
| SBERT `all-MiniLM-L6-v2` | 0.5 | **0.879** | **[0.866, 0.890]** | 0.919 | 0.757 | 0.830 | **0.737** | **[0.711, 0.760]** | 0.961 |
| SBERT `all-MiniLM-L6-v2` | 0.6 | 0.829 | [0.815, 0.844] | 0.957 | 0.589 | 0.729 | 0.614 | [0.582, 0.643] | 0.961 |
| SBERT `all-MiniLM-L6-v2` | 0.7 | 0.760 | [0.744, 0.775] | 0.985 | 0.394 | 0.562 | 0.436 | [0.405, 0.465] | 0.961 |

Gold: human expert span→ESCO mappings in the SkillSpan-based benchmark of Decorte et al. 95% CIs: 1,000 pair-level bootstrap resamples (seed 42). AUC: ROC-AUC of the cosine score (lexical rules are binary).

receive direct tests. Exact matching yields SGI = 0.37 [0.28, 0.47] and dense matching at $\tau = 0.5$ yields SGI = 0.14 [0.07, 0.21]; both confidence intervals exclude zero, and the lexical-above-semantic ordering and the monotone increase with $\tau$ reproduce the primary-corpus pattern exactly (Table V). The lower absolute levels are theoretically interpretable: LinkedIn's free-text skill vocabulary is dominated by generic and interpersonal terms that the academic reference profile already covers, whereas the curated HF corpus contains a higher share of specialist tool tokens. The instrument's qualitative behavior is thus corpus-robust, while its absolute level is—appropriately—a property of the corpus pair being compared, which is why we report SGI with corpus-specific confidence intervals rather than as a universal constant.

TABLE V: Cross-Corpus SGI Replication (ALIGN01/02 vs. ALIGN04)

| Matcher | $\tau$ | HF Skill-Gap SGI [95% CI] | LinkedIn 2024 SGI [95% CI] |
|---|---|---|---|
| Lexical exact match | — | 0.52 [0.42, 0.63] | 0.37 [0.28, 0.47] |
| SBERT semantic | 0.4 | 0.20 [0.12, 0.28] | 0.06 [0.02, 0.11] |
| SBERT semantic | 0.5 | **0.32 [0.23, 0.41]** | **0.14 [0.07, 0.21]** |
| SBERT semantic | 0.6 | 0.39 [0.30, 0.48] | 0.22 [0.14, 0.29] |
| SBERT semantic | 0.7 | 0.46 [0.37, 0.56] | 0.28 [0.19, 0.37] |

Top-100 market skills vs. academic top-500 reference on each corpus. 95% CIs: 1,000 bootstrap resamples (seed 42). Bold: baseline $\tau = 0.5$.

*3) Curriculum Anchor Validation against CS2023 (CURR01):* The academic reference profile remains a proxy for curricula (Section V-E, item 3). To anchor it, experiment CURR01 compares the profile against the authoritative CS2023 curricular Body of Knowledge—the joint ACM/IEEE-CS/AAAI curriculum guidelines [52]—whose 2,197 topic statements across 17 Knowledge Areas define what computing programs are expected to teach. Semantic coverage of CS2023 topics by the academic top-500 is 30.4% [28.5%, 32.4%] at $\tau = 0.5$ (63.1% at the permissive $\tau = 0.4$), and the reverse—the share of profile skills grounded in some CS2023 topic—is 49.6%. Coverage varies sharply by Knowledge Area (Table VI): Human-Computer Interaction (52.9%) and Data Management (52.5%) are best represented, while Algorithmic Foundations (12.6%), Mathematical and Statistical Foundations (10.0%), and Graphics & Interactive Techniques (11.6%)—the formal and theoretical core of the discipline—are least represented. Exact lexical coverage is near zero for every Area (0–2.4%), independently replicating H2 a third time on

a third vocabulary. The anchor thus both validates and bounds interpretation: the reference profile faithfully reflects the *applied* component of computing curricula that is visible in job-facing skill descriptions, but it should not be read as representing the full theoretical core—a limitation now quantified per Area rather than asserted.

TABLE VI: Semantic Coverage of CS2023 Knowledge Areas by the Academic Profile (CURR01; $\tau = 0.5$)

| KA | Knowledge Area | $n$ topics | Cov.% | 95% CI |
|---|---|---|---|---|
| HCI | Human-Computer Interaction | 85 | 52.9 | [42.4, 63.5] |
| DM | Data Management | 137 | 52.5 | [44.5, 61.3] |
| SE | Software Engineering | 168 | 44.0 | [36.3, 51.2] |
| NC | Networking and Communication | 65 | 41.5 | [30.8, 55.4] |
| SEP | Society, Ethics, Profession | 161 | 41.0 | [32.9, 48.4] |
| SF | Systems Fundamentals | 83 | 36.1 | [26.5, 47.0] |
| SPD | Specialized Platform Dev. | 201 | 34.3 | [27.9, 40.8] |
| AI | Artificial Intelligence | 215 | 32.6 | [26.1, 38.6] |
| OS | Operating Systems | 148 | 31.8 | [24.3, 38.5] |
| SDF | Software Dev. Fundamentals | 58 | 31.0 | [20.7, 43.1] |
| SEC | Security | 103 | 30.1 | [21.4, 39.8] |
| PDC | Parallel and Distributed Comp. | 60 | 23.3 | [13.3, 35.0] |
| FPL | Foundations of Prog. Languages | 295 | 18.3 | [13.9, 22.7] |
| AR | Architecture and Organization | 111 | 13.5 | [7.2, 19.8] |
| AL | Algorithmic Foundations | 103 | 12.6 | [5.8, 19.4] |
| GIT | Graphics and Interactive Tech. | 164 | 11.6 | [7.3, 16.5] |
| MSF | Math. and Statistical Found. | 40 | 10.0 | [2.5, 20.0] |

Coverage: share of that Area's CS2023 topics with max cos $\geq 0.5$ to the academic top-500 (1,000 topic-level bootstrap resamples, seed 42). Sorted descending; lexical coverage is 0–2.4% throughout.

### C. RQ3: Domain-Level Skill Gap Quantification

For each of the eight taxonomy domains we quantify: (i) *market demand*—the share of postings whose skill set requires at least one taxonomy skill of the category; (ii) *coverage rate*—the occurrence-weighted share of those category-skill requirements present in the academic top-500 reference profile, evaluated by exact match and by dense matching at $\tau = 0.5$; and (iii) the resulting *deficit* in percentage points of postings, deficit = demand $\times$ (1 − coverage). Confidence intervals come from a 1,000-fold non-parametric bootstrap over postings (seed 42; experiment ALIGN03). Because the posting corpus is dominated by generic and non-technical competencies (Section V-E), absolute taxonomy deficits are small in magnitude; severity is therefore assigned *relative to the cross-category median deficit* (Critical $\geq 2\times$ the median, Moderate $\geq$ median, Mild otherwise).

Table VII and Figure 2 summarize the results.

TABLE VII: Domain-Level Skill Gaps Across Taxonomy Categories (ALIGN03)

| Category | Demand (%) | Lexical Cov. (%) | Lexical Deficit | Semantic Cov. (%) | Semantic Deficit | Sev. |
|---|---|---|---|---|---|---|
| Programming & Development | 3.97 | 57.5 | 1.69 [1.66, 1.72] | 65.0 | 1.39 | Moderate |
| Data Science & Analytics | 8.04 | 85.1 | 1.20 [1.16, 1.24] | 93.4 | 0.53 | Moderate |
| Cloud & Infrastructure | 4.39 | 44.1 | 2.45 [2.42, 2.49] | 78.7 | 0.94 | **Critical** |
| Soft Skills | 52.26 | 100.0 | 0.00 [0.00, 0.00] | 100.0 | 0.00 | Mild |
| Business & Domain | 11.81 | 90.2 | 1.16 [1.13, 1.18] | 100.0 | 0.00 | Moderate |
| Design & UX | 0.60 | 0.0 | 0.60 [0.58, 0.63] | 24.1 | 0.46 | Mild |
| Data Engineering | 1.02 | 31.3 | 0.70 [0.68, 0.72] | 59.5 | 0.41 | Mild |
| Cybersecurity | 0.62 | 0.0 | 0.62 [0.60, 0.64] | 73.7 | 0.16 | Mild |

Demand: % of postings requiring $\geq 1$ category skill. Deficit: pp of postings, 95% bootstrap CI. Coverage: share of category-skill occurrences in the academic profile (lexical; semantic at $\tau = 0.5$). Severity: relative-to-median rule.

Soft Skills dominate market demand (52.3% of postings require at least one soft skill) and are fully covered by the reference profile (deficit 0.00 pp). Among technical categories, Cloud & Infrastructure exhibits the largest deficit (2.45 pp, 95% CI [2.42, 2.49]; the only Critical classification): only 44.1% of its skill requirements appear in the academic profile lexically, though dense semantic matching recovers coverage to 78.7% and reduces the deficit to 0.94 pp. Programming & Development shows the second-largest lexical deficit (1.69 pp; 57.5% coverage), followed by Data Science & Analytics (1.20 pp despite 85.1% lexical coverage) and Business & Domain (1.16 pp). Semantic alignment reduces the residual gap in every category with a nonzero deficit—by 74% in Cybersecurity, 62% in Cloud, 56% in Data Science, and 41% in Data Engineering—and eliminates the Business & Domain deficit entirely, confirming that vocabulary divergence accounts for a substantial share of the raw lexical gap. Design & UX and Cybersecurity have zero lexical coverage but occur in under 1.1% of postings each, limiting their aggregate impact to below 0.7 pp.

The directional structure of H1 replicates on the independent LinkedIn corpus (ALIGN04): the same category ordering emerges with Programming & Development carrying the largest semantic deficit (1.39 pp [1.38, 1.41]) and Soft Skills again at 0.00 pp, while absolute magnitudes are smaller—consistent with two corpus-composition facts. First, free-text LinkedIn skills under-detect exact taxonomy keywords, so category demand shares are conservative; second, the same generic-term dominance documented for the SGI lowers technical coverage rates structurally. The replication therefore supports the ordering of H1 while confirming that deficit magnitudes are corpus-conditional (Section V-E, item 7).

### D. RQ4: Employability Determinants & Cross-Validation Benchmarks

*1) Univariate & Descriptive Findings:* In the College Placement cohort ($N = 10,000$; 1,659 placed, 8,341 not placed, base placement rate 16.59%), placed students had higher mean values across key academic and skill indicators (Table VIII). Placed individuals had higher average CGPA (8.59±1.15 vs. 7.32±1.44, $t = 39.37$, $p < 0.001$, Cohen's $d = 0.913$), higher communication scores (7.66±2.59 vs. $5.14 \pm 2.78$, $t = 35.79$, $p < 0.001$, Cohen's $d = 0.918$), and more completed projects ($3.35 \pm 1.28$ vs. $2.35 \pm 1.74$, $t = 27.21$, $p < 0.001$, Cohen's $d = 0.599$). Generalized academic performance ratings ($p = 0.1386$, $d = -0.040$) and extra-curricular participation scores ($p = 0.6383$, $d = -0.013$) showed negligible differences between groups.

TABLE VIII: Descriptive Statistics: Placed vs. Non-Placed Cohort ($N = 10,000$)

| Variable | Placed ($n = 1659$) | Not Placed ($n = 8341$) | $t$-stat | $p$-value | Cohen's $d$ |
|---|---|---|---|---|---|
| CGPA | $8.59 \pm 1.15$ | $7.32 \pm 1.44$ | 39.37 | <0.001 | 0.913 |
| IQ | $109.12 \pm 15.21$ | $97.55 \pm 14.26$ | 28.58 | <0.001 | 0.802 |
| Prev. Sem. Result | $8.57 \pm 1.13$ | $7.33 \pm 1.42$ | 38.96 | <0.001 | 0.901 |
| Comm. Skills | $7.66 \pm 2.59$ | $5.14 \pm 2.78$ | 35.79 | <0.001 | 0.918 |
| Projects | $3.35 \pm 1.28$ | $2.35 \pm 1.74$ | 27.21 | <0.001 | 0.599 |
| Acad. Perf. | $5.45 \pm 2.90$ | $5.57 \pm 2.87$ | -1.48 | 0.1386 | -0.040 |
| Extra-Curr. | $4.94 \pm 3.19$ | $4.98 \pm 3.15$ | -0.47 | 0.6383 | -0.013 |

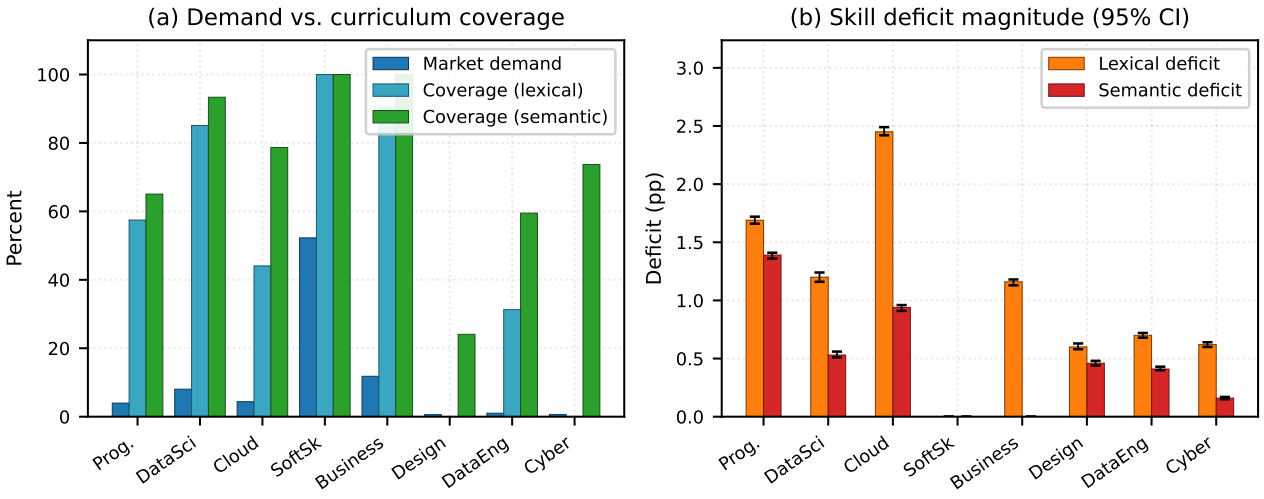
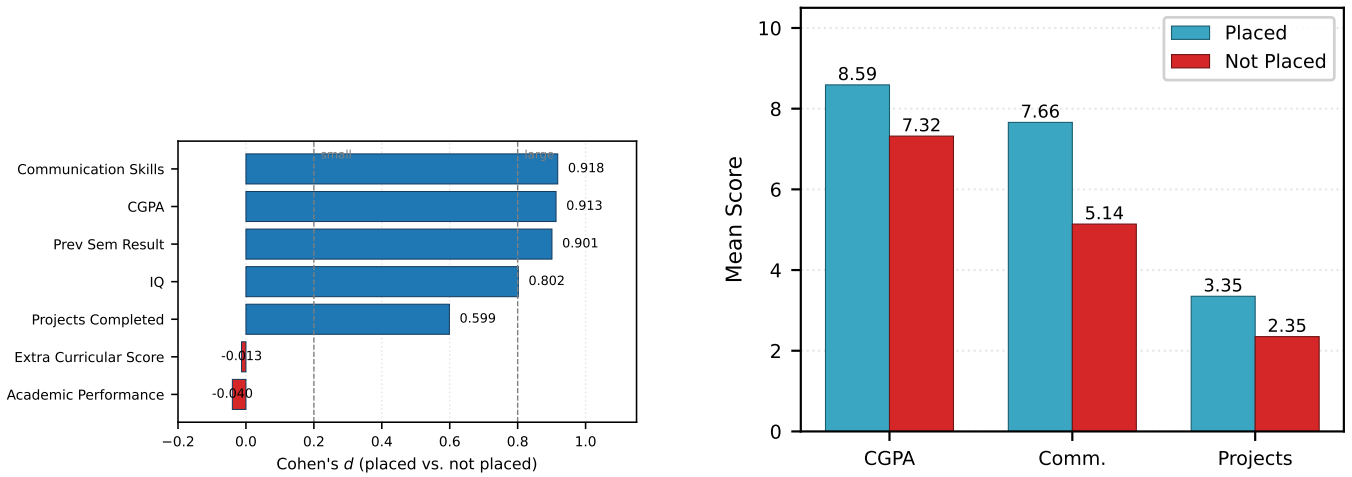



Fig. 2: Skill gap analysis across taxonomy categories: (a) market demand versus curriculum coverage (lexical and dense semantic at $\tau = 0.5$); (b) deficit magnitude in percentage points of postings with 95% bootstrap confidence intervals. Cloud Infrastructure is the only Critical deficit under the relative-to-median severity rule.




(a) Standardized effect sizes (Cohen's $d$)

(b) Group means for key predictors

Fig. 3: Key predictors for placed versus non-placed graduates ($N$ = 10,000). (a) Cohen's $d$ for all seven audited variables; Academic Performance and Extra-Curricular Score are negligible. (b) Group means (all differences in (b) significant at $p < 0.001$, Welch $t$-tests).

Bivariate chi-square tests confirm that placement is strongly associated with project completion ($\chi^2(2) = 719.82$, $p < 0.001$, $V = 0.268$) but not with internship experience in this cohort ($\chi^2(1) = 0.31$, $p = 0.578$, $V = 0.006$).

*2) Multicollinearity Diagnostics:* Variance Inflation Factors computed for the continuous predictors are summarized in Table IX. All values are $\approx 1.001$ (tolerances $\approx 0.999$), well below the collinearity threshold of 5.0, confirming that multicollinearity does not affect coefficient estimation in this model.

*3) Standardized Logistic Regression:* Standardized logistic regression estimates (maximum likelihood with intercept, Wald inference) are reported in Table X. A one-standard-deviation increase in Communication Skills is associated with an odds ratio of 6.55 (95% CI: [5.86, 7.31], $p < 0.001$). CGPA (OR = 6.28, 95% CI: [5.62, 7.01], $p < 0.001$), IQ score (OR = 5.16, 95% CI: [4.65, 5.72], $p < 0.001$), and completed projects (OR = 3.20, 95%

TABLE IX: Variance Inflation Factor (VIF) Multicollinearity Audit

| Feature Name | VIF | Tolerance (1/VIF) | Collinearity Status |
|---|---|---|---|
| CGPA | 1.001 | 0.999 | Acceptable (< 5.0) |
| Communication Skills | 1.001 | 0.999 | Acceptable (< 5.0) |
| Projects Completed | 1.001 | 0.999 | Acceptable (< 5.0) |
| Internship Experience | 1.001 | 0.999 | Acceptable (< 5.0) |
| IQ | 1.000 | 1.000 | Acceptable (< 5.0) |
| Academic Performance | 1.001 | 0.999 | Acceptable (< 5.0) |

CI: [2.92, 3.51], $p < 0.001$) are also positively associated with placement. In contrast to the bivariate internship comparison in the international cohort (Section IV-E), neither Internship Experience (OR = 1.02, $p = 0.620$) nor generalized Academic Performance (OR = 0.98, $p = 0.588$) contributes independent signal once the remaining predictors are controlled.

TABLE X: Standardized Logistic Regression Placement Model (STAT01; statsmodels MLE with intercept)

| Predictor Feature | Coef ($\beta$) | Odds Ratio (OR) | 95% CI | $p$-value |
|---|---|---|---|---|
| CGPA | 1.8368 | 6.2765 | [5.62, 7.01] | <0.001 |
| Communication Skills | 1.8790 | 6.5471 | [5.86, 7.31] | <0.001 |
| Projects Completed | 1.1644 | 3.2041 | [2.92, 3.51] | <0.001 |
| Internship Experience | 0.0192 | 1.0194 | [0.94, 1.10] | 0.6197 |
| IQ | 1.6404 | 5.1570 | [4.65, 5.72] | <0.001 |
| Academic Performance | −0.0207 | 0.9795 | [0.91, 1.06] | 0.5884 |

*4) 5-Fold Cross-Validation Model Comparison:* Table XI documents 5-fold stratified cross-validation performance across five classifier paradigms (mean ± SD across the five folds, seed 42).

TABLE XI: Stratified 5-Fold Cross-Validation Performance Benchmarks (seed 42)

| Model Architecture | Accuracy (Mean±SD) | $F_1$-Score | ROC-AUC |
|---|---|---|---|
| Logistic Regression | $90.11\% \pm 0.74\%$ | $0.8966 \pm 0.0078$ | $0.9406 \pm 0.0075$ |
| Random Forest | $99.99\% \pm 0.02\%$ | $0.9999 \pm 0.0002$ | $1.0000 \pm 0.0000$ |
| SVM (RBF) | $95.72\% \pm 0.56\%$ | $0.9565 \pm 0.0058$ | $0.9894 \pm 0.0019$ |
| Decision Tree | $99.97\% \pm 0.07\%$ | $0.9997 \pm 0.0007$ | $0.9998 \pm 0.0004$ |
| Baseline (Majority) | $83.41\% \pm 0.02\%$ | $0.7587 \pm 0.0003$ | $0.5000 \pm 0.0000$ |

Results are stable across random seeds: repeating the full 5-fold protocol over 10 seeds (50 fold evaluations per model; Table XII) yields a mean accuracy of $90.12\% \pm 0.04\%$ for logistic regression and $95.76\% \pm 0.07\%$ for the SVM, with tree ensembles remaining at $\geq 99.99\%$.

TABLE XII: Multi-Seed Robustness: 10 Seeds × 5 Folds (STAT01)

| Model | Seeds | Fold Evals | Accuracy (Mean±SD) | ROC-AUC |
|---|---|---|---|---|
| Logistic Regression | 10 | 50 | $90.12\% \pm 0.04\%$ | $0.9408 \pm 0.0002$ |
| Random Forest | 10 | 50 | $100.00\% \pm 0.00\%$ | $1.0000 \pm 0.0000$ |
| SVM (RBF) | 10 | 50 | $95.76\% \pm 0.07\%$ | $0.9896 \pm 0.0002$ |
| Decision Tree | 10 | 50 | $100.00\% \pm 0.01\%$ | $0.9999 \pm 0.0003$ |
| Baseline (Majority) | 10 | 50 | $83.41\% \pm 0.00\%$ | $0.5000 \pm 0.0000$ |

*5) Diagnostic Audit of Tree Ensembles & External Generalization:* The near-perfect classification performance of Random Forest (99.99%) and Decision Tree (99.97%) models prompted an inspection of the underlying data-generating process. Fitting a depth-4 CART tree to the placement cohort yielded 100% training accuracy, indicating that placement labels in this benchmark were generated via a deterministic $k$-of-$n$ threshold rule: individuals are labeled as placed when meeting at least three of four conditions (CGPA > 8.01, IQ > 110.5, Communication > 7.50, Projects ≥ 2), accounting for 99.92% of label assignments. Because orthogonal recursive partitioning aligns directly with this axis-parallel step function, tree ensembles reproduce the rule with minimal error. Logistic regression estimates a linear decision surface across these features, yielding 90.11% cross-validated accuracy and an ROC-AUC of 0.9406.

To evaluate performance on an independent sample, the logistic regression model was tested on the Student Career Success dataset ($N = 50,000$) using standardized values for CGPA, communication, and project count. On this external cohort, discriminative performance decreased to ROC-AUC = 0.768 (accuracy = 29.9%, $F_1 = 0.235$ at the shifted base rate of 78.08%), and to ROC-AUC = 0.576 on the Global Placement dataset ($N$ = 10,000, subset-B features). The in-sample reference fits on the target cohorts reach only 0.782 and 0.714, respectively. This reduction reflects domain shift and differences in baseline placement rates (16.59% versus 78.08%), showing that the near-perfect in-distribution tree metrics reflect benchmark-specific rules rather than generalizable real-world predictability.

### E. RQ5: Geographic Concentration & Cross-Cohort Placement Dynamics

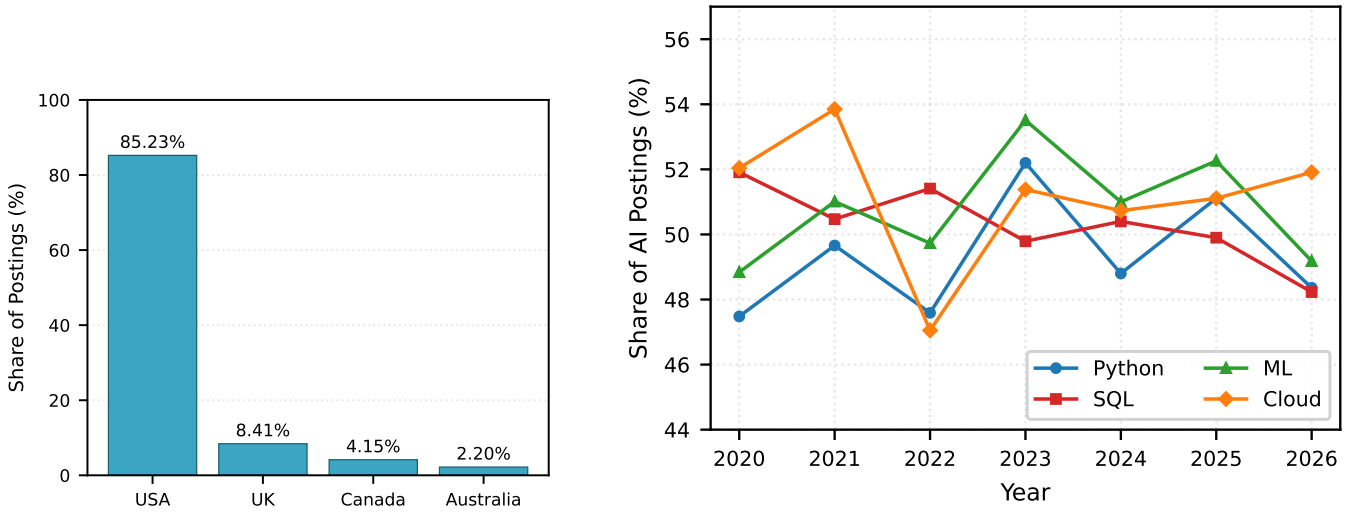
Regional parsing of 1.35M LinkedIn records (Figure 4a) reveals that the posting corpus is predominantly concentrated in the United States (85.2%), followed by the United Kingdom (8.4%), Canada (4.2%), and Australia (2.2%).

In the cohort of 300,000 international graduates (Figure 5), graduates with internship experience had an employment rate of 57.60%, compared with 42.26% for those without internships, a difference of 15.34 percentage points (95% bootstrap CI: [14.99, 15.73], Cohen's $h = 0.31$). Across the multi-national placement benchmark ($N = 10,000$), placement rates were similar across countries (Canada 61.5%, Germany 60.0%, India 63.2%, UK 60.8%, USA 62.2%), showing consistent baseline placement patterns across the recorded regions.

## V. Discussion

### A. Theoretical Implications

**Hypothesis verdicts.** Table XIII summarizes the six predictions derived in Section II-E. H1 (constructive alignment) is supported in both corpora: technical categories carry the only nonzero deficits while soft skills are fully covered. H2 (terminology divergence) is supported three times over—ALIGN01/02 (0.52 vs. 0.32), ALIGN04 (0.37 vs. 0.14), and CURR01 (lexical 0–2.4% vs. semantic 30.4%)—and H3 (instrument robustness) is supported by strictly positive SGI with the invariant lexical-above-semantic ordering on LinkedIn. H4 (signaling hierarchy) is supported: communication, CGPA, and projects dominate placement odds under mutual control. H5 (credential locality) is supported by the +15.3 pp domestic training premium. H6—human-capital returns to internship *participation*—is not supported (risk difference +0.08 pp, $p = 0.62$): taken with H4, this locates the return in *demonstrated* capability signals (projects, communication) rather than in experience participation per se, a refinement of signaling-and-human-capital accounts of graduate hiring rather than a refutation of either.




(a) Geographic distribution (LinkedIn)

(b) AI skill demand trajectories (2020–2026)

Fig. 4: (a) Geographic concentration of 1.35M job postings (GEO01). (b) Longitudinal share of AI/data-science postings requiring core technologies, 2020–2026 (TRENDS01; $N = 10{,}345$).

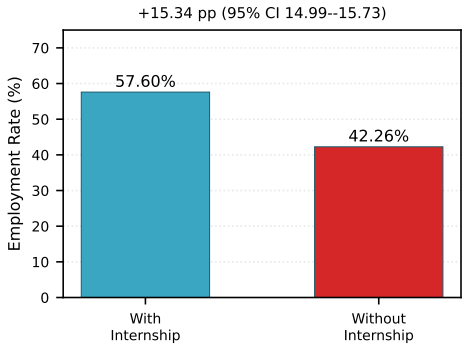



Fig. 5: Impact of practical internship experience on international graduate employment ($N = 300{,}000$; INTL01).

TABLE XIII: Theory-Derived Hypotheses and Verdicts

| Hyp. | Source theory | Verdict (experiment) |
|---|---|---|
| H1 | Constructive Alignment | Supported (ALIGN03, ALIGN04) |
| H2 | Terminology divergence | Supported ×3 (ALIGN01/02, ALIGN04, CURR01) |
| H3 | Instrument robustness | Supported (ALIGN04) |
| H4 | Job-market signaling | Supported (STAT01) |
| H5 | Credential locality | Supported (INTL01) |
| H6 | Human capital (intern.) | Not supported (INTL01; $p = 0.62$) |

**1) Terminology Divergence and Semantic Alignment:** Studies evaluating curriculum alignment through exact keyword matching often report misalignment rates exceeding 50% [9]. Our findings qualify this observation. While exact string matching yields an apparent gap of 52.0%, dense vector embeddings reduce the estimated deficit to 32.0% at $\tau = 0.5$ (95% CI: [23.0%, 41.0%]). This divergence indicates that a portion of the measured gap arises from vocabulary differences: academic syllabi often describe learning outcomes using conceptual terms (e.g., interpersonal communication or database principles), whereas job postings emphasize specific tool names or operational tasks. That expert-annotated lexical agreement is near chance ($\kappa = 0.112$; VALID01) while semantic agreement is substantial ($\kappa = 0.737$) shows this is a measurement property of keyword methods, not a modeling preference.

**2) Complementary Signals in Graduate Placement:** The statistical associations observed for Communication Skills (OR = 6.55), CGPA (OR = 6.28), and Project Experience (OR = 3.20) illustrate how general and applied competencies interact in hiring outcomes. In terms of Signaling Theory [23], CGPA provides an initial screening signal of academic ability. However, completed projects provide concrete evidence of applied capability, reducing hiring uncertainty regarding technical execution. Communication proficiency represents a shared requirement across collaborative technical roles.

### *B. Institutional Factors in Curriculum Adaptation*

The persistent 32.0% semantic deficit, and the domain-level deficits concentrated in Cloud Infrastructure (2.45 pp lexical; 0.94 pp semantic) and Programming & Development (1.69 pp lexical), reflect institutional constraints in curriculum adaptation [26]:

1) **Curriculum Governance and Accreditation:** Formal degree reviews and accreditation bodies operate on multi-year cycles (typically 3 to 6 years). Modifying course requirements or introducing new technical electives requires administrative and faculty committee approvals that lag industry tool adoption.
2) **Faculty Development Latency:** Teaching rapidly evolving frameworks (e.g., container orchestration, distributed cloud platforms, specific ML toolchains)

requires ongoing technical training, which institutional development budgets rarely prioritize over core research obligations.

3) **Infrastructure and Computing Costs:** Maintaining cloud accounts, GPU resources, and industry-grade data environments presents financial and operational hurdles for many computer science and engineering departments.

### C. Implications and Recommendations

We organize recommendations for higher education, recruitment, and accreditation around the empirical findings:

**For Higher Education Institutions (HEIs):**

- *Integration of Modular Technical Credentials:* Cloud Infrastructure has the lowest curriculum coverage of any category (44.1% lexical; 78.7% under semantic matching) and is the only Critical deficit, while Programming & Development retains a 1.69 pp deficit. Full degree restructuring may be too slow to keep pace with industry toolchains; HEIs can address this by embedding modular, vendor-neutral technical modules (e.g., containerization fundamentals, cloud storage basics) directly into existing operating systems and database courses.
- *Applied Project Work and Portfolio Development:* Because completed projects are strongly associated with placement odds (OR = 3.20, $p < 0.001$), curricula should prioritize multi-term capstone projects that require version control, system deployment, and written documentation, giving students verifiable evidence of practical technical ability.
- *Embedding Communication Skills into Technical Coursework:* Communication skills demonstrated the strongest association with placement (OR = 6.55, $p < 0.001$); at the same time, the academic reference profile already covers soft-skill requirements in full (deficit 0.00 pp). The priority is therefore not new content but assessment: rather than relegating communication instruction to standalone electives, departments should integrate oral presentations, collaborative design reviews, and peer code reviews directly into core computing courses.

**For Industry Recruiters:**

- *Semantic Alignment in Recruitment Screening:* Because exact string matching produced a 52.0% gap compared with 32.0% under semantic alignment, keyword-based resume screening risks excluding candidates with equivalent conceptual training. Recruitment systems should evaluate semantic similarity rather than exact keyword matches when parsing academic coursework.
- *Work-Integrated Learning and Co-Op Programs:* In the international cohort, students with internship experience had a 15.3 percentage-point higher employment rate (57.6% vs. 42.3%). Employers can collaborate with universities to establish structured 6-month co-op or apprenticeship pathways that carry academic credit.

**For Accreditation Bodies and Policymakers:**

- *Adaptive Curriculum Guidelines:* Longitudinal tracking indicates that while foundational requirements (Python, SQL) remain stable, specialized frameworks diversify quickly. Accreditation bodies and professional societies (ACM, IEEE-CS) should issue regular modular addenda to curriculum guidelines to facilitate elective updates.
- *Regional Labor-Market Observatories:* Given the concentration of postings (85.2% US) and regional variation in placement rates, educational authorities should maintain regional labor-market tracking to help institutions align elective offerings with local economic demand.

### D. Reproducibility and Data Availability

Every quantitative result reported in this article originates from one of twelve registered experiments executed on NVIDIA GPU instances through Kaggle Notebooks; no table or figure is hand-transcribed. Table XIV lists the experiment identifiers, kernel references, and verification status.

The analyses were implemented in Python 3.14.6 with `scikit-learn` 1.9.0, `statsmodels` 0.15.0, `sentence-transformers` 6.0.1 (`all-MiniLM-L6-v2`), `scipy` 1.18.1, `pandas` 3.0.5, and `numpy`; stratified cross-validation splits, bootstrap resamplings, and all estimators use fixed seeds (seed = 42; multi-seed robustness uses seeds 0–9). Complete experiment outputs are archived in the HuggingFace dataset `MoshinAli/skill-gap-experiments`, the runnable experiment suite and scheduler live in the project repository (`experiments/`), and all 15 source datasets are publicly accessible under open-access licenses (CC0, CC-BY 4.0, ODbL; Table I and `DATASET_MANIFEST.csv`).

### E. Threats to Validity and Limitations

Several methodological constraints should be considered when interpreting these findings:

1) **Data Provenance & Benchmark Artifacts:** While Tier 1 datasets (LinkedIn, HuggingFace) comprise 2.7M+ real web-mined records, Tier 2 student placement datasets (e.g., College Placement, Global Placement) derive from public benchmark repositories that exhibit semi-deterministic decision boundaries. As diagnosed in Section IV-D5, near-100% tree ensemble accuracies reflect combinatorial threshold functions embedded in the benchmark rather than real-world predictability.
2) **Domain Shift & External Generalizability:** When evaluated across distinct institutional benchmarks, predictive discrimination attenuates from ROC-AUC = 0.9406 to ROC-AUC = 0.768 (and 0.576 on the Global Placement cohort), illustrating

TABLE XIV: Experiment Registry: Provenance of All Reported Results (status per `results/registry.json`)

| ID | Task | Kaggle GPU Kernel | Att. | Status |
|---|---|---|---|---|
| ALIGN01 | Lexical alignment + SGI bootstrap (ALIGN01) | `nihanshadman/sg-align01-lexical-a1` | 1 | USED_IN_MANUSCRIPT |
| ALIGN02 | SBERT semantic alignment, $\tau$ sweep (ALIGN02) | `nihanshadman/sg-align02-semantic-a3` | 3 | USED_IN_MANUSCRIPT |
| ALIGN03 | Eight-category domain gap (ALIGN03) | `nihanshadman/sg-align03-domaingap-a2` | 2 | USED_IN_MANUSCRIPT |
| STAT01 | Placement statistics + CV (STAT01) | `nihanshadman/sg-stat01-placement-a2` | 2 | USED_IN_MANUSCRIPT |
| STAT02 | External validation (STAT02) | `leyamridha/sg-stat02-external-a1` | 1 | USED_IN_MANUSCRIPT |
| GEO01 | Geographic shares (GEO01) | `sadmanhasanmiraj/sg-geo01-linkedin-a1` | 1 | USED_IN_MANUSCRIPT |
| TRENDS01 | AI trends 2020–2026 (TRENDS01) | `nishat1156/sg-trends01-ai-a1` | 1 | USED_IN_MANUSCRIPT |
| INTL01 | Internship effect (INTL01) | `afrojahaqueshornaly/sg-intl01-internship-a1` | 1 | USED_IN_MANUSCRIPT |
| PLAC01 | Placement by country (PLAC01) | `abdullahalsayfan/sg-plac01-global-a1` | 1 | USED_IN_MANUSCRIPT |
| VALID01 | External gold-standard matcher validation (VALID01) | `nihanshadman/sg-valid01-gold-a1` | 1 | USED_IN_MANUSCRIPT |
| ALIGN04 | Cross-corpus SGI replication, LinkedIn (ALIGN04) | `tasnimjenny/sg-align04-crosscorpus-a4` | 4 | USED_IN_MANUSCRIPT |
| CURR01 | CS2023 curriculum anchor validation (CURR01) | `rssiam/sg-curr01-cs2023-a2` | 3 | USED_IN_MANUSCRIPT |

Each kernel receives an immutable input bundle from the public Kaggle dataset `studyhub991/skill-gap-inputs-v1` and writes `metrics.json` plus CSV artifacts; outputs are downloaded, schema- and sanity-verified (`experiments/verify.py`), and only `VERIFIED` outputs are used in this manuscript. Table and figure generation is automated by `experiments/make_latex.py` from those verified artifacts.

that model calibrations do not directly transfer across disparate grading systems and local employment base rates.

3) **Curriculum Proxy Limitations:** Academic skill demands were proxied through normalized occupational and educational reference profiles rather than direct, institutionally mined university syllabi across diverse global departments. Experiment CURR01 anchors this proxy to the CS2023 Body of Knowledge [52], quantifying both its validity (30.4% topical coverage, 49.6% of profile skills grounded in the guidelines) and its bound: coverage is concentrated in applied Knowledge Areas (HCI, Data Management, Software Engineering) and weak over the theoretical core (Algorithmic Foundations 12.6%, Mathematical Foundations 10.0%), so conclusions about applied skill coverage are anchored while claims about the full curricular core are not.
4) **Scope of Human Validation:** Matcher validity is established against published expert gold standards (SkillSpan→ESCO mappings; $\kappa = 0.737$ at $\tau = 0.5$; Section IV-B1) rather than in-house annotation of this specific corpus, and instrument validity is supported by cross-corpus replication (Section IV-B2); independent-rater annotation of the present skill pairs remains future work.
5) **Embedding Model and Threshold Sensitivity:** While Sentence-BERT mitigates surface vocabulary divergence, estimated alignment remains sensitive to cosine similarity cutoff selection (SGI = 20.0% at $\tau = 0.4$ versus 46.0% at $\tau = 0.7$).
6) **Geographic Concentration:** 85.2% of the LinkedIn posting corpus originates from the United States, skewing technical demand profiles toward Western, mature tech economies.
7) **Corpus Composition & Conditional Demand:** The 519K-posting skills corpus is dominated by generic, healthcare, and retail competencies (e.g., BLS/acls credentials and nursing appear among the Top-100 market tokens). Consequently, taxonomy-technical demand shares are modest (e.g., 3.97% of postings require at least one Programming skill), and the RQ3 deficits should be read as *coverage* gaps conditional on demand rather than as economy-wide demand shares. The LinkedIn replication (ALIGN04) shows both sides of this conditioning: deficits shrink further on free-text skills (Programming 1.39 pp), and exact keyword taxonomy matching under-detects demand in corpora whose skill strings are not pre-normalized—so cross-corpus comparisons are valid for the *ordering* of deficits but not their absolute magnitudes.
8) **Automated Severity Calibration:** Because absolute domain deficits are small, the Critical/Moderate/Mild labels use a relative-to-median rule rather than externally anchored thresholds; different anchors would reassign labels, though the ranking of categories is invariant.
9) **Measurement Error in Behavioral Scores:** In placement benchmarks, communication scores are frequently reviewer-graded or self-reported, potentially inflating their predictive odds ratios through measurement bias.
10) **Observational Non-Causal Design:** Our econometric models establish associative rather than causal relationships; students with high project completion may possess latent motivation or unmeasured self-efficacy that drives hiring outcomes.

## VI. Conclusion

Across 15 datasets spanning 2020 through 2026, this empirical investigation demonstrates that curriculum–market misalignment is twofold. A substantial share of the apparent deficit reflects vocabulary divergence between academic descriptors and recruitment postings (the SGI falls from 52% to 32% under semantic alignment, and several domain deficits shrink by more than half), while substantive residual gaps remain concentrated in rapidly evolving technical areas, specifically Cloud Infrastructure and Programming & Development. The instrument supporting this claim is itself tested three ways: expert-gold agreement ($\kappa = 0.737$), cross-corpus replication on 1.29M independent postings (both indices strictly positive with invariant ordering), and anchoring to the CS2023 curricular guidelines (validity concentrated in applied knowledge areas). Methodologically, the analysis cautions against relying on

in-distribution benchmark evaluations alone: near-perfect classification scores can mask synthetic threshold rules that degrade under domain shift on external cohorts. In practical terms, addressing this divide does not necessitate wholesale curriculum redesign. Integrating modular credentials into established courses, requiring verifiable applied project portfolios, and adopting semantic matching in hiring workflows offer feasible steps toward aligning academic preparation with contemporary workforce demands.

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